NearBNB

Internal MVP Product & Technical Requirements

Internal working document for product, backend, frontend, QA, DevOps and implementation planning. This document captures the detailed decisions and flows discussed while converting the client material into a scalable MVP.

1. Source and Decision Approach

The client-supplied document is treated as a source of product intent, not as an unquestionable technical specification. The corrected draft and subsequent discussions establish the working product model. The most important correction is that NearBNB uses exact timestamp availability and flexible check-in with 24-hour-multiple stays, rather than a standard calendar-night model.

The original client draft describes a standard accommodation marketplace and includes listing, search, booking, payment, cancellation, payout, reviews, administration and audit requirements. The corrected requirements adapt those concepts to the time-based booking model.

2. Core Product Model

Concept	Working decision
Marketplace	Three-sided marketplace: Guest, Host, Admin.
Launch market	Australia-first, but multi-country ready.
Currency	Australia-first configuration; currency must be configurable for future countries.
Booking type	Request-to-book only for MVP.
Check-in	Flexible time.
Check-in interval	30 minutes initially; system should allow configuration for 15/30/60-minute intervals.
Duration	Exact multiples of 24 hours only.
Inventory	Property contains separate bookable categories/sub-listings.
Availability	Exact timestamp ranges.
Payment	Stripe.
Marketplace payout	Stripe Connect.

Host payout eligibility	Configurable delay after check-in; initial value 24 hours.
Refund	Platform-controlled; normal cancellations are automatic, exceptional host refunds require platform/admin handling.
Reviews	After completed stay, guest reviews host/property and host reviews guest.
Disputes	Complex dispute module deferred from MVP.

3. Core Entities and Relationship

User

↓

Host Profile (when approved)

↓

Property

↓

Category / Sub-listing

↓

Availability

↓

Booking

↓

Payment

↓

Transfer / Host Payout

A Property is the physical accommodation. A Category/Sub-listing is the actual independently bookable inventory unit or room type. Price and availability belong to the category/sub-listing. A Booking references the category/sub-listing and exact check-in/check-out timestamps.

- User: common authentication and profile information.
- Host Profile: host verification status, payout status and Stripe connected-account reference.
- Property: address, location, physical property information and host relationship.
- Category/Sub-listing: bookable unit/type, price, capacity, amenities and category-specific information.
- Availability: open/blocked time ranges for the category/sub-listing.
- Booking: guest, category, exact time range, guest count, pricing snapshot and state.

- Booking Hold: temporary reservation of an exact time range while a request/payment is pending.
- Guest Verification: booking-level guest identity/verification details where required.
- Payment: payment provider transaction and status.
- Transfer/Payout: host amount, transfer status and Stripe payout information.
- Refund: refund request and provider result.
- Review: completed-booking feedback.
- Policy Version: immutable cancellation/fee/tax/business-rule snapshot.
- Audit Event: important action history.

4. Registration and Authentication Flow

Registration means creating a NearBNB account. Email verification confirms that the email belongs to the user.

1. Guest/host applicant enters registration information.
2. Account is created in an unverified state.
3. System sends email verification OTP/link.
4. User verifies email.
5. User can log in and use allowed features.
6. Server-side role permissions are applied to every protected endpoint.
7. Password reset, logout and session expiry are supported.

5. Host Onboarding and Activation

Host activation should use multiple gates. Admin approval alone should not make a host fully active for booking operations.

8. User registers a normal NearBNB account.
9. User verifies email.
10. User selects Become a Host.
11. Host application information is collected.
12. Required identity/host verification is completed.
13. Application is submitted for admin review.
14. Admin approves the host.
15. Host completes Stripe Connect onboarding for payout capability.
16. Stripe connected-account status is checked.
17. Only when all mandatory activation conditions are satisfied is the host marked Active.
18. An active host can create/submit properties and use the host dashboard.

Recommended activation rule: host must be approved by admin AND have required verification completed AND have an acceptable Stripe Connect payout status. If any required condition is missing, the host remains Pending/Restricted and cannot receive confirmed bookings.

6. Stripe Connect Host Setup

NearBNB should not collect and store the host's bank details in its own normal host form solely for payout processing. The host should be sent through Stripe Connect onboarding.

19. Host opens Payout Setup in NearBNB.
20. NearBNB creates or retrieves the host's Stripe Connected Account.
21. NearBNB creates a Stripe-hosted onboarding/account-link flow.
22. Host completes required identity/business and payout information on Stripe.
23. Stripe verifies/updates the account status.
24. NearBNB stores the Stripe connected-account identifier and relevant non-sensitive status fields.
25. NearBNB prevents booking activation if required payout capability is not ready.

The exact Stripe Connect account type and configuration should be finalized during implementation based on the platform's legal entity, country, Stripe account country and marketplace charge/transfer model.

7. Property and Category/Sub-listing Flow

26. Active host creates a property.
27. Host enters property information, address, capacity and other required details.
28. Host creates one or more categories/sub-listings.
29. Each category has its own price, capacity, details and availability.
30. Host submits the property/listing for admin approval.
31. Admin approves, rejects or requests changes.
32. Only approved and active categories/sub-listings appear in public search.
33. Material changes can trigger re-approval.

Example:

Property: ABC Resort
↓
Standard Category
↓
Premium Category

↓
Pool Villa

Each category can have independent availability and price.

8. Guest Search and Availability Flow

34. Guest enters location, check-in date/time and guest count.
35. System finds approved active categories/sub-listings that can accommodate the guest.
36. System checks whether the requested check-in time plus the selected minimum duration can fit within open availability.
37. Guest opens a property/category page.
38. Guest can open the availability view.
39. Guest selects a check-in date/time and duration.
40. System immediately returns Available or Not Available.
41. If unavailable, UI explains the conflicting period and can show alternative available periods.

9. Availability UX Options

The implementation can combine multiple UI methods. Recommended MVP approach is a combined calendar + selected-day timeline + direct availability selector.

Method	Purpose	Example
Calendar overview	Quick understanding of which dates are free, partially occupied or unavailable.	Green/amber/red date indicators.
Selected-day time view	Understand exact occupied/open time ranges.	25 Sep 2 PM → 26 Sep 2 PM booked.
Timeline	Visually show long booking blocks across exact time ranges.	Open → booked → open.
Direct availability selector	Let the guest choose the actual booking without interpreting the calendar.	Check-in 25 Sep 2 PM + 24 hours → Available.
Combined approach	Best overall UX for MVP.	Calendar → time view → check-in → duration → availability result.

The UI should use simple messages such as: “Someone is staying from 25 Sep 2:00 PM until 26 Sep 2:00 PM.” Users should not be required to understand the internal timestamp/range logic.

10. Booking Request Flow

42. Guest selects category/sub-listing.
43. Guest selects check-in timestamp.
44. Guest selects 24/48/72-hour duration.
45. System validates capacity, listing status and availability.
46. Guest enters required booking/verification information.
47. System creates a pending booking request and a temporary hold on the exact requested range.
48. Other guests cannot request the held period while the hold is active.
49. Host receives the request.
50. Host accepts or rejects.
51. If rejected, hold is released.
52. If the request expires without host action, hold is released.
53. If accepted, guest is notified to pay.

Open value: host response deadline must be configured before final implementation.

11. Booking Status Model

Status	Meaning
Draft	Guest started booking but request/hold not completed.
Requested / Pending	Booking request exists and exact slot is temporarily held.
Accepted / Awaiting Payment	Host accepted; guest must complete payment.
Payment Pending	Payment is being processed and hold remains active.
Confirmed	Payment successful; exact time range permanently blocked.
Checked In	Check-in timestamp has passed and stay has started.
Completed	Checkout timestamp has passed.
Cancelled	Booking cancelled by guest, host or admin.
Refund Pending	Refund has been initiated and provider result is pending.
Refunded	Refund completed.
Rejected / Expired	Host rejected or request/payment expired; hold released.

Failed	Payment failed or other terminal booking failure.

12. Payment Flow

54. Host accepts the booking request.
55. NearBNB creates the payment flow for the guest.
56. Guest pays the total through Stripe.
57. NearBNB does not store raw card details.
58. Stripe processes the payment.
59. Stripe webhook/event confirms the payment result.
60. Payment record is updated idempotently.
61. On successful payment, booking becomes Confirmed.
62. The exact time range becomes permanently unavailable.
63. Guest and host receive confirmation.

Important: payment success should be finalized from the provider event/webhook rather than trusting only the browser redirect.

13. Financial Model

The system must distinguish between guest payment, platform revenue/fees, host amount, transfer and bank payout.

Guest
↓
Stripe payment
↓
NearBNB marketplace financial record
↓
Host transfer
↓
Host Stripe Connected Account
↓
Stripe payout
↓
Host bank account

A Stripe transfer to a connected account and the later payout from Stripe to the host's external bank account are different events and should be tracked separately.

Example only:

Component	Example
Accommodation	AUD 200.00
Cleaning fee	AUD 20.00
Guest/platform service fee	AUD 15.00
Applicable tax	AUD 23.50
Guest total	AUD 258.50
Platform commission	Configured separately
Host amount	Calculated from configured financial rules

14. Country, Currency, Tax and Commission Design

Australia is the starting market, but the platform must be designed for additional countries. Country-specific financial configuration should not be hard-coded in application branches.

- Country.
- Region/state where relevant.
- Currency.
- Tax configuration.
- Platform commission.
- Guest service fee.
- Other applicable fees.
- Rounding rules.
- Tax responsibility/reporting configuration where required.

The booking must store the actual financial values used at booking time so later configuration changes do not alter historical bookings.

Exact tax rates, tax responsibility and legal reporting obligations require confirmation with the client's tax/legal adviser for each launch country.

15. Cancellation and Refund Flow

Normal guest cancellation:

Guest

↓

Cancel booking

↓

System checks policy

↓

Refund calculated

↓

Guest confirms

↓

Booking cancelled

↓

Stripe refund

↓

Refund status recorded

Host cancellation:

Host

↓

Cancel booking

↓

Booking cancelled

↓

Applicable guest refund

↓

Host cancellation recorded

↓

Admin visibility/flag

Exceptional host refund request:

Host

↓

Refund request

↓

Admin/platform review

↓

Approve or reject

↓

Stripe refund if approved

↓

Financial records updated

Normal policy-based guest cancellation should not require manual admin or host approval. The system should automatically calculate and initiate the applicable refund.

16. Refund and Stripe Reconciliation

- Every refund must reference the original payment.
- Support full and, if required by the business rules, partial refunds.
- Record refund amount, reason, initiator, approver, provider reference and timestamps.
- If host funds have already been transferred, the implementation must reconcile the refund with the related host transfer according to the selected Stripe Connect charge/transfer model.
- Refund operations must be idempotent and webhook/provider results must be recorded.

17. Payout Flow

64. Booking becomes Confirmed after successful guest payment.
65. NearBNB calculates the host payable amount from the booking's stored financial snapshot.
66. Guest reaches the configured check-in time.
67. Payout eligibility timer starts.
68. Initial recommended release delay is 24 hours after check-in.
69. When eligible, NearBNB initiates the host transfer according to the selected Stripe Connect model.
70. Transfer status is stored.
71. Stripe then pays the connected account according to its payout configuration/schedule.
72. NearBNB records payout status and provider references.

The exact payout schedule can also depend on Stripe's connected-account capabilities and country configuration. Our business rule should control eligibility, while Stripe controls the supported external payout mechanics.

18. Guest Verification

- Required guest information should be captured according to the final client-approved verification policy.

- The earlier corrected requirements included full name, age, email, phone and at least one approved social profile.
- If social verification remains required, the platform should validate the URL format and approved platform domain.
- Sensitive guest information must not be public.
- Host visibility should follow the approved visibility rule; current working decision is full verification details become visible after booking confirmation.
- Admin has platform-wide visibility for support/fraud handling.

19. Reviews

73. Booking reaches Completed.
74. Review period starts.
75. Guest can review host/property.
76. Host can review guest.
77. One review per party per booking.
78. Property/host reviews can contribute to public property rating.
79. Host-to-guest reviews are not public property reviews.
80. For future bookings, the host may be shown a guest trust summary according to the final privacy/visibility rules.
81. Admin can moderate reviews.

Recommended MVP visibility: public guest-facing pages show property/host reviews; host-to-guest review comments remain private to relevant hosts/admins, with a possible guest trust summary exposed to a host when appropriate.

20. Disputes — Post-MVP

A full dispute-management system is deferred. MVP should not include a complex evidence, escalation and case-management module.

- Optional basic issue/report entry.
- Admin/support can handle urgent issues manually.
- Full dispute case management is future scope.

21. Guest Capabilities

- Register and verify email.
- Login/logout/password reset.

- Manage own profile.
- Search and filter.
- View property/category details.
- View availability.
- Create booking requests.
- View request status.
- Pay accepted requests.
- View confirmed/past bookings.
- Cancel according to policy.
- View receipts/refunds.
- Review completed stays.
- Raise basic issues.

22. Host Capabilities

- Register/login.
- Apply to become a host.
- Complete host verification.
- Connect Stripe payout account.
- Manage own properties.
- Manage own categories/sub-listings.
- Manage prices and availability.
- Submit listings for approval.
- Accept/reject booking requests.
- View own reservations.
- View guest information for own bookings according to visibility rules.
- View own transaction, refund, earnings and payout information.
- Request exceptional refunds.
- Review guests.
- Report issues.

Host must not have platform-wide user management, other hosts' financial visibility, unrestricted refund authority or control over platform configuration.

23. Admin Capabilities

Admin Module	Capabilities
Dashboard	Platform overview and operational alerts.

Users	Search/view users, account status, relevant booking history, suspend/reactivate.
Hosts	Applications, verification status, approve/reject/request changes, suspend/reactivate.
Properties	Review, approve/reject/request changes, deactivate/unpublish.
Categories	View/manage platform category configuration where applicable.
Bookings	Search/view all bookings, statuses, dates, guests, financial links and permitted operational actions.
Payments	View platform-wide payment records and provider references.
Refunds	View, approve/process permitted refunds and see provider outcome.
Payouts	View host payable amounts, transfers, payout status and failures.
Reviews	Moderate/hide reviews and record reason.
Configuration	Fees, commissions, cancellation policies, verification requirements and agreed business settings.
Audit Logs	Review important administrative/system actions.
Admin Profile	Name, email, password, 2FA status and security/session information.

24. Audit Log

Audit logs are a permanent record of important actions. They answer who performed an action, what changed, when it happened and why.

Example action	Recorded information
Property approved	Admin, property, action, timestamp, reason.
Host suspended	Admin, host, action, timestamp, reason.
Refund approved	Admin, booking, amount, reason, timestamp, provider reference.
Booking modified	Actor, booking, before/after values, reason, timestamp.
Host cancellation	Host, booking, reason, timestamp.

25. Admin Profile and Security

- Name.
- Email.
- Password change.
- 2FA enable/disable and status.
- Last login/session information.
- Security events should be recorded through the audit/security history rather than creating a large activity section inside the profile.

26. Notifications

- Email verification.
- Password reset.
- Host application status.
- Listing submission and approval/rejection.
- Booking request.
- Host acceptance/rejection/expiry.
- Payment success/failure.
- Booking confirmation.
- Cancellation/refund status.
- Check-in reminder.
- Checkout reminder.
- Review invitation.

27. Permissions

Action	Guest	Host	Admin
Manage own profile	Yes	Yes	Yes
Create property	No	Yes	Controlled/admin-assisted if needed
Approve property	No	No	Yes
Accept booking request	No	Own properties	Admin override where defined
View own bookings	Yes	Own properties	All
View guest verification	Own details	Own confirmed bookings	All
View own transactions	Yes	Yes	Yes

View other hosts' transactions	No	No	Yes
Process normal cancellation	Own booking	Own cancellation	Permitted admin action
Exceptional refund	Request/support	Request	Approve/process
Manage users globally	No	No	Yes
Manage platform fees	No	No	Yes
Moderate reviews	No	No	Yes

28. Non-Functional Requirements

- Secure authentication and password hashing.
- Server-side authorization on every protected operation.
- TLS in production.
- Rate limiting and protected secrets.
- No raw card storage.
- UTC storage for timestamps with local-time display where appropriate.
- Money stored as integer minor units rather than floating-point values.
- Database transactions and overlap constraints to prevent double booking.
- Idempotency for payment/refund/webhook processing.
- Soft deletion/status preservation for transaction-linked records.
- Structured logging without sensitive guest/payment data.
- Monitoring, health checks, backups and restore testing.
- Pagination for large admin/search lists.
- Responsive web experience.

29. Double-Booking Protection

The availability engine must be server-side and transactional. A UI availability check is not enough.

82. Search checks current availability.
83. Booking request creates a temporary hold for the exact time range.
84. The database prevents conflicting confirmed/active holds according to the defined state rules.
85. Before payment confirmation, availability is revalidated.

86. Booking confirmation and inventory blocking occur atomically.
87. Concurrent booking attempts cannot both become confirmed for overlapping time on the same category/sub-listing.

30. Price and Policy Snapshot

A booking must keep the exact price, fees, tax values and applicable cancellation/policy version used at booking time. Later changes to configuration must not change historical bookings.

- Accommodation amount.
- Applicable fees.
- Tax amount and relevant tax configuration reference.
- Platform fee/commission values.
- Discounts/adjustments if used.
- Final payable amount.
- Host payable amount.
- Cancellation policy version.
- Currency.

31. MVP vs Future Scope

MVP	Future
Responsive web	Native iOS/Android apps
Request-to-book	Instant Book if later required
24-hour multiples	More complex duration/pricing models
Exact timestamp availability	Advanced inventory/pricing optimization
Stripe + Connect	Additional payment providers
Basic issue reporting	Full dispute/case management
Basic reviews	Advanced trust/reputation scoring
Single-language launch	Multi-language
Australia-first, multi-country-ready architecture	Full multi-country operations/tax workflows
Basic admin configuration	Advanced business rules/configuration engine
Email notifications	SMS/push/in-app notification expansion
Basic search/filtering	AI ranking/recommendations

32. Client Decisions Still Required

- Maximum stay duration.
- Final check-in interval: 15, 30 or 60 minutes; current recommendation 30 minutes.
- Booking request expiry period.
- Host identity-verification method.
- Required guest verification fields and accepted social platforms, if this requirement remains.
- Platform commission.
- Guest service fee.
- Tax responsibility, rates and legal treatment for Australia and future countries.
- Exact host payout release delay; current recommendation 24 hours after check-in.
- Stripe Connect country/account configuration and final marketplace charge/transfer model.
- Cancellation policy templates and refund treatment.
- Guest trust/review visibility rules.
- Whether saved properties are required in MVP.
- Whether guest-host messaging is required in MVP.

33. Suggested Implementation Sequence

Phase	Outcome
1. Foundation	Architecture, environments, authentication, roles, database foundation, audit framework, media storage.
2. Host & Inventory	Host onboarding, Stripe setup, property/category creation, approval and availability.
3. Guest Discovery	Search, filters, property/category details and availability UX.
4. Booking	Request/hold/accept/reject/expiry and booking state machine.
5. Payment	Stripe payment, webhook processing, confirmation and financial snapshots.
6. Cancellation/Refund/Payout	Policy engine, refund flow, Stripe reconciliation and host payout flow.
7. Dashboards	Guest, host and admin dashboards.
8. Reviews/Notifications	Completed-stay reviews and email notifications.
9. Hardening	Security, concurrency tests, monitoring, backups, restore and production readiness.

34. Definition of Done

The MVP is complete when the complete transaction works in a controlled production booking: host registration and approval, Stripe payout setup, property/category approval, exact-time availability, booking request and hold, host acceptance, guest payment, confirmed booking, stay completion, configured payout, cancellation/refund handling and post-stay reviews. Admins must be able to operate the platform without direct database intervention, and critical payment/booking concurrency and recovery scenarios must be tested.

35. Important Implementation Notes

- Do not model the core availability engine as calendar-night inventory.
- Do not use floating-point values for money.
- Do not store raw payment-card data.
- Do not trust only frontend availability checks.
- Do not confirm a booking only because the browser returned from payment.
- Do not allow unrestricted host refunds.
- Do not mix host transactions with platform-wide admin financial visibility.
- Do not hard-code Australia-only tax/commission logic into business code.
- Do not hard-delete financial/booking history.
- Keep provider IDs, payment, refund, transfer and payout records linked for reconciliation.

36. Reference Basis

This internal document is based on the client-supplied NearBNB MVP requirements, the corrected property-booking MVP requirements, and the subsequent product decisions recorded in the project discussion. The source material establishes the core listing, search, booking, payment, cancellation, payout, review, admin, data-integrity and non-functional requirements, while the later decisions resolve the time-based booking, category/sub-listing, request-to-book, refund and payout questions.